

Beyond Imperfections: Exploring Defects for Breakthroughs in Perovskite Solar Cell Research

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The field of solar systems has undergone rapid development with the emergence of the special material, perovskite. Perovskite's unique mechanism, defect tolerance, has enabled perovskite solar cells (PSCs) to achieve high power conversion efficiencies (PCEs), and many studies on this subject have been conducted. "Defect tolerance" indicates that the defects in perovskite are primarily generated at the shallow-energy level and do not occur through nonradiative recombination. However, this also indicates that defects are well formed in perovskite films and that the shallow defects can transform into deep traps, leading to long-term stability issues. Therefore, controlling defects in perovskite is essential for developing PSCs with high PCEs. The causes of defects in perovskite are diverse, and the defect patterns differ considerably, particularly depending on the location of the PSCs. In this review, the defects generated in perovskite will be discussed and review several methods for passivating them at different.

the most well-known HTM, Spiro-OMeTAD.^[11] Dopants (e.g., Li-TFSI and t-butylpyridine) are required to achieve adequate hole mobility and, in turn, high PCEs. However, these dopants are hygroscopic; as such, they can deteriorate the perovskite by absorbing moisture and by decreasing the thermal stability of the HTM by lowering its glass transition temperature (T_g).^[11]

In the quest for nontoxic alternatives to lead, researchers have developed tin-based perovskites,^[12] double perovskites,^[13] and other perovskite derivatives. However, these alternatives still suffer from low efficiency and stability,^[14] which require substantial efforts to overcome. Additionally, various deposition methods have been proposed for the large-area processing

of perovskite,^[15] such as antisolvent-free deposition,^[16] inkjet printing,^[17] blade coating,^[18] and slot-die coating.^[19] However, reproducibility and crystal uniformity remain key issues that need to be addressed.

Regarding the PCE, PSCs have developed at a very rapid pace, and are currently showing decent PCE that is comparable to that of silicon solar cells that have been successfully commercialized. However, there is still substantial room for improvement as it still falls short of the ideal PCE of 33% achieved by a semiconductor with an optical bandgap of 1.4 eV (the S-Q limit).^[20] This disparity signifies that despite the considerable efficiency achieved, there remains significant scope for improvement. In many PSCs currently delivering high efficiencies, the reported short-circuit current density (J_{sc}) values are close to 25 mA cm⁻², almost reaching the limit.^[21] Conversely, the open circuit voltage (V_{oc}) still shows a large deviation from the S-Q limit value of 1.32 V under AM 1.5G sunlight illumination. This indicates that to increase the PCE of PSCs, we need to focus on improving the V_{oc} . Although many studies have been conducted to solve the current challenges, more methods should be developed.

In 2014, Snaith, and coworkers, for the first time, reported "the defect passivation method," which achieves material passivation by filling the trap sites in perovskite.^[22] They found that there are many uncoordinated halide ions on the perovskite crystal surface, which trap positive charges, and reduce efficiency. They employed iodopentafluorobenzene (IPFB) for passivation through halogen bonding with the trap state, which led to an increase in efficiency.

1. Introduction

Over the last decade, significant efforts have been devoted to the advancement of perovskite solar cells (PSCs), achieving high power conversion efficiencies (PCEs) of up to 26.0%.^[1] This achievement is mainly attributed to the valuable properties of hybrid organic-inorganic perovskite materials, such as their high absorption coefficients,^[2] long diffusion lengths,^[2,3] low exciton binding energies,^[4] and high mobility.^[5] Consequently, PSCs can be promising alternatives to conventional silicon solar cells, which have high processing costs and heavy weights.^[6] Currently, the most core research parts in the field of perovskite solar cells are as follows: (1) poor stability under real operating conditions^[7] (e.g., humidity, light, and temperature), (2) the toxicity of lead,^[8] (3) poor large-area processability,^[9] and (4) low efficiency compared to the Shockley-Queisser limit (S-Q limit).

Attempts to solve some of these challenges are in progress.^[10] Various dopant-free hole transporting materials (HTM) have been developed to partially overcome the critical limitations of

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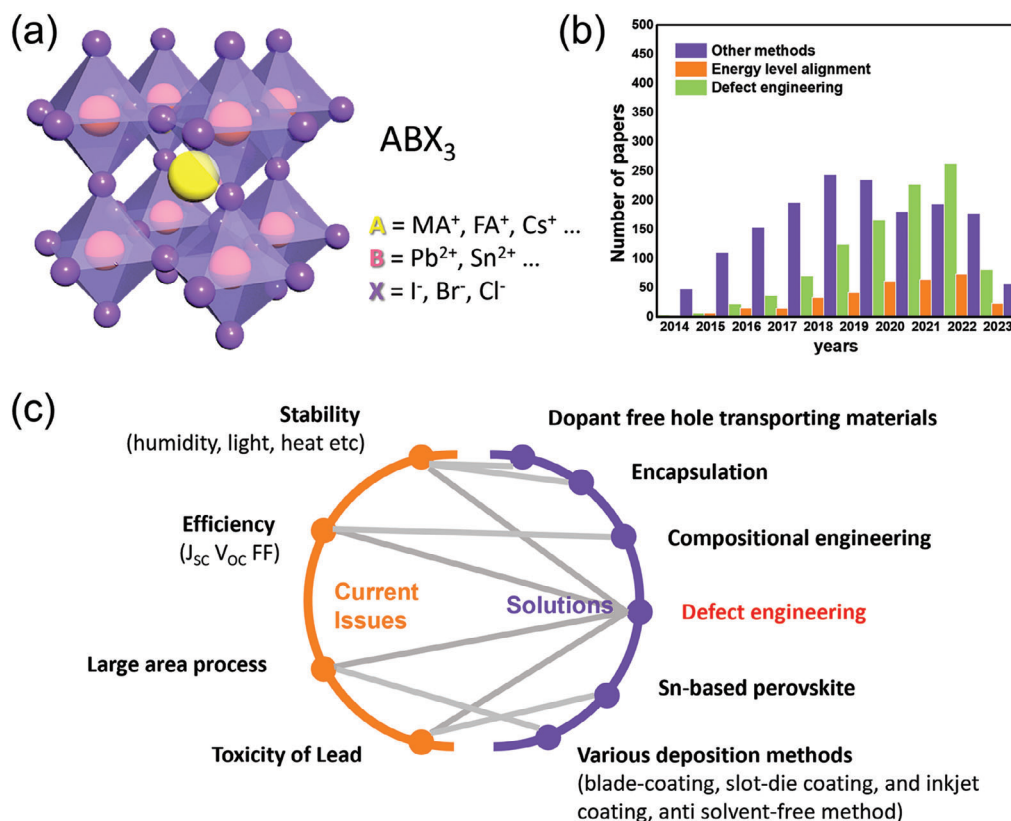


Figure 1. a) Schematic illustration of the perovskite structure. b) Number of papers related to V_{OC} enhancement for PSCs. c) Schematic diagram of the current challenges affecting PSCs and their solutions.

Defects on perovskites are one of the main reasons for the significant deviation of the V_{OC} from the S–Q limit and why the corresponding devices exhibit instability under operational conditions, particularly at elevated temperatures.^[23] Defects can be uncontrollable in large-area processing, causing uniformity problems.^[24] Therefore, the development of technologies for minimizing defects in the preparation processes and retarding defect formation during operation is paramount for improving efficiency and stability simultaneously. Further, this approach may contribute to the development of large-area devices with excellent quality and uniformity.

Figure 1b depicts the number of studies each year focusing on improving the V_{OC} of PSCs. According to the data, defect engineering has become highly important, occupying over 50% of the employed approaches in 2022. Defect engineering holds potential as a solution to improve the overall perovskite performance and address various challenges. Consequently, it is crucial to organize and analyze the existing defect engineering research systematically.

In this review, we delve into the understanding of perovskite defects and how they influence the PCE of PSCs. Additionally, we classify the defects located at each layer in the device. In each location of the device, different aspects of the problem caused by the defects manifest. Therefore, we address the defects in three parts and proffer solutions to each of them: (1) the hole transport layer (HTL)/perovskite interface, (2) electron transport layer (ETL)/perovskite interface, and (3) bulk perovskite. Finally, we ad-

dress the current limitations of PSCs from the viewpoint of defect engineering.

2. What are Defects in Perovskite Crystals?

Perovskite's notable advantage lies in its capacity for low-temperature processing in solution, owing to its low formation energy. However, this low formation energy also hinders the formation of highly crystalline perovskite films at low temperatures, resulting in the formation of low-quality polycrystalline films. These films exhibit numerous defects at the grain boundaries between crystals, with a significantly higher defect density compared to that of a single perovskite crystal.^[25]

Perovskites exhibit a periodic crystal lattice structure known as the ABX_3 structure, and certain disordered regions within this structure act as defects.^[26] Perovskite is composed of three elements, making it somewhat more complex with various types of defects compared to crystals composed of one or two elements. Defects are categorized into various types based on the cause of their formation and structure. For example, vacancy defects occur when an ion is missing from a lattice site, interstitial defects arise when an ion occupies an interstitial site, and substitutional defects occur when an original ion is replaced by another ion at a lattice site (see **Figure. 2**). Notably, perovskites have many dangling bonds—bonds with unsatisfied valence—on their surface or grain boundaries, which are also considered defects. Additionally, the iodine vacancy (V_I) captures electrons, leading to

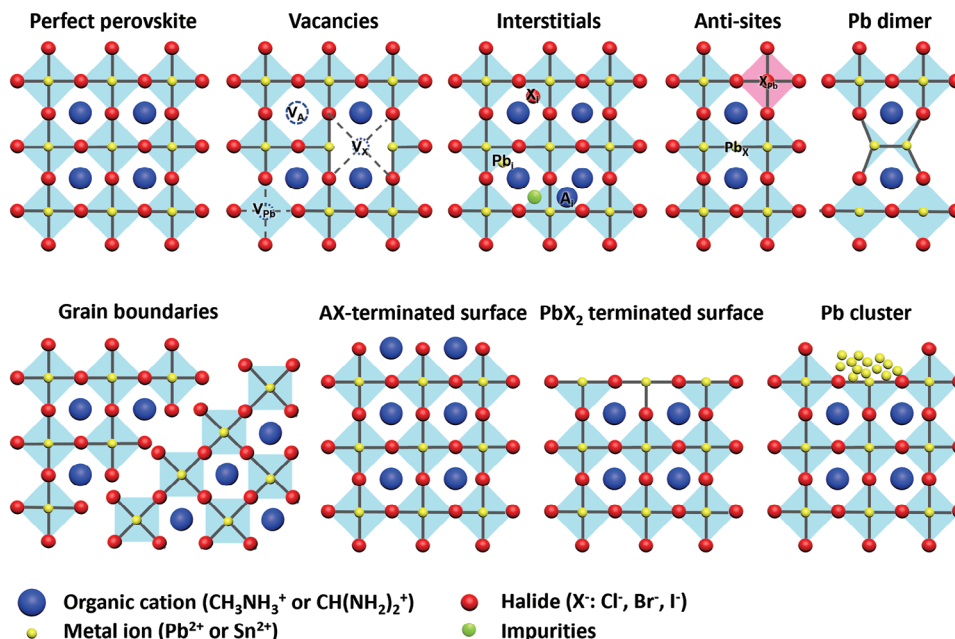


Figure 2. Various type of perovskite defects.

two Pb atoms moving toward each other and interacting to form a Pb dimer, which is another type of defect. Thermal degradation can further lead to the formation of Pb clusters, also categorized as defects.^[27] These defects serve as traps for charge carriers with different energy levels and can significantly affect the performance of the corresponding device.

Various defects exert differing effects on the optoelectrical properties of the device, depending on their energy levels. Therefore, it is crucial to carefully examine the energy levels of these defects present in perovskite.^[28] As you can see, the energy level of vacancy defects is primarily located near the conduction band minimum (CBM) and valence band maximum (VBM), while anti-site defects are situated midway between the CBM and VBM. Consequently, the impacts of vacancy defects and anti-site defects on devices are markedly distinct (this will be discussed in detail later). The formation energy of these defects varies depending on whether the perovskite is enriched or deficient in each element (see Figure 3b). In other words, the formation energy can vary significantly depending on the crystal growth conditions, which implies that under specific crystal growth conditions, a relatively high concentration of defects can be formed.^[29]

Furthermore, certain defects lead to unintentional doping. V_{MA} , V_{Pb} , I_i , MA_{Pb} , I_{MA} , and I_{Pb} act as acceptor defects, while V_i , MA_i , Pb_i , Pb_{MA} , MA_i , and Pb_i serve as donor defects. To avoid unintentional doping, it is imperative to maintain stoichiometric growth conditions. In such cases, Schottky disorder occurs, where equal numbers of cation and anion vacancies compensate for each other by carrying opposite charges.^[30]

3. Impact of Defects on Perovskite Solar Cells

As mentioned in the introduction section, the reason why PSCs have not yet reached the S–Q limit is that their V_{OC} shows a large

deviation from the S–Q limit. V_{OC} is highly related to the quasi-Fermi level splitting (QFLS). When perovskite is irradiated with light, electrons are excited from the conduction band (CB) to the valence band (VB). In the equilibrium state, electrons, and holes each have new Fermi levels that are different from the original Fermi levels. This is called QFLS, and the quasi-Fermi level of the electron is $E_{F,e}$, whereas the quasi-Fermi level of the hole is $E_{F,h}$ (Figure 4a).

Ideally, the V_{OC} of a PSC is related to the difference between the quasi-Fermi levels when the perovskite is irradiated with light.

$$-q V_{OC} = E_{F,e} - E_{F,h} \quad (1)$$

However, sub-gap states resulting from defects reduce the QFLS.^[31] Therefore, defect passivation is a strategy that can improve the V_{OC} by suppressing the decrease in the QFLS due to the defective sub-gap state.^[32] For instance, deQuilettes et al. reported that surface passivation using n-trioctylphosphine oxide increased the QFLS.^[33]

The device V_{OC} can be also calculated as follows:

$$V_{OC} = V_{OC}^{rad} - V_{OC}^{non-rad} \quad (2)$$

Nonradiative recombination that can occur in semiconductors includes defect-assisted recombination (Shockley–Read–Hall (SRH) recombination), Auger recombination, band-tail recombination, and interface recombination. Among these, auger recombination, and band-tail recombination are known to exert little effect on the PCE of PSCs.^[21]

The reason why perovskite solar cells do not meet the S–Q limit is because the S–Q limit is a function of the semiconductor bandgap and only considers loss due to light absorption and cooling of the hot carrier. However, not only the loss due to cooling of the hot carrier but also the loss due to SRH

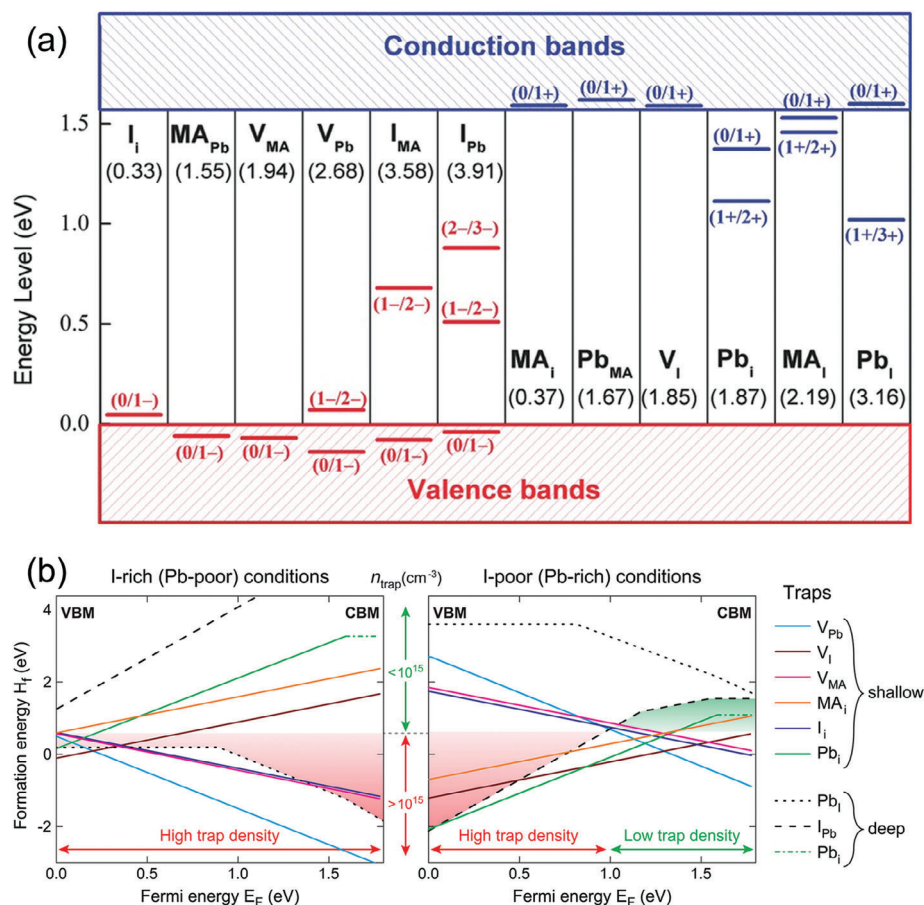


Figure 3. a) Formation energies of neutral defects of perovskite.^[28] Copyright 2014, Wiley-VCH. b) Formation energy of defects in perovskite grown in I-rich conditions and I-poor conditions.^[29] Copyright 2014, American Chemical Society.

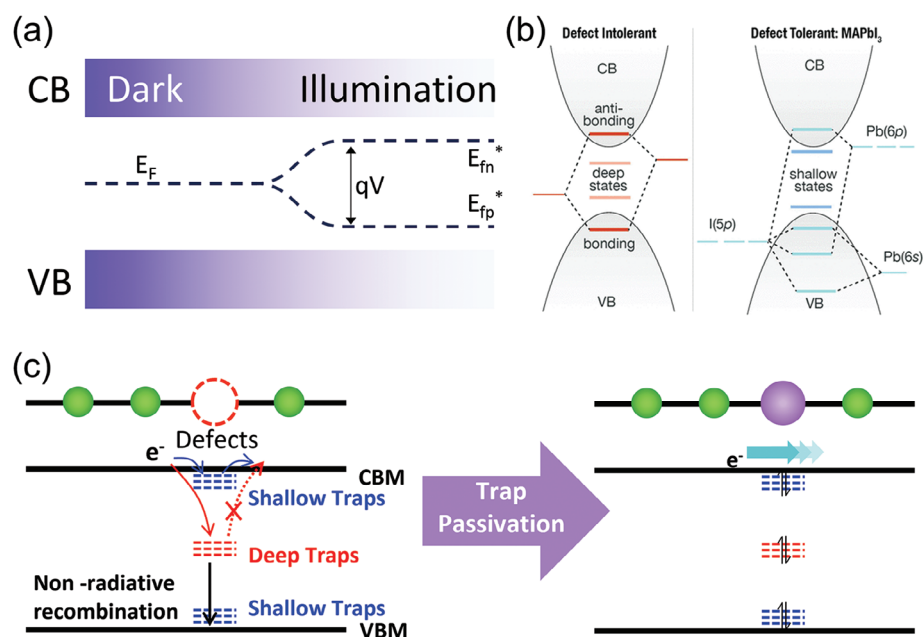


Figure 4. a) Schematic illustration of the quasi-Fermi level splitting (QFLS). b) Schematic illustrations of the defect tolerance of the perovskite crystals. Reproduced with permission.^[36] Copyright 2017, American Chemical Society. c) Schematic illustration of the shallow traps and deep traps in the perovskite crystal and trap passivation.

recombination appears to be a very dominant loss not only in perovskite but also in all semiconductors. Therefore, it is very difficult to reach the S-Q limit, and SRH recombination becomes a very important factor in determining efficiency.^[34] SRH recombination is caused by defects. Defects can damage the device's performance by acting as traps for photogenerated free carriers. However, not all perovskite defects seriously affect the efficiency of PSCs.

Defects in perovskite are divided into shallow traps and deep traps. The criterion for their categorization is the energy-level difference (ΔE) between the trap's energetic location and the band edge. If this ΔE is less than the thermal energy ($k_B T$, where k_B is Boltzmann's constant and T is temperature) (i.e., $\Delta E < k_B T$), a shallow trap is generated. Carriers caught in a shallow trap can easily escape by absorbing thermal energy. Conversely, if ΔE is larger than $k_B T$ (i.e., $\Delta E > k_B T$), a deep trap is generated. Carriers caught in a deep trap cannot escape by absorbing thermal energy. Instead, they undergo nonradiative recombination, i.e., SRH recombination.^[35]

Fortunately, most defects in perovskite have shallow energy-level states; therefore, carriers trapped in these shallow traps do not promote nonradiative recombination. This is the unique "defect tolerance" feature of perovskites.

Defect tolerance is a unique property that does not occur in conventional semiconductors, such as CdTe, CuInSe₂, and Cu₂ZnSnS₄. Such unique defect properties are attributed to the strong antibonding coupling between the s orbital of Pb and the p orbital of I, which causes most atomic orbitals to be in the CB or VB of the crystal (Figure 4b).^[36] Consequently, when bonds are broken, the resulting orbitals are mostly inside the bands or close to the band edges, rather than in the middle of the bandgap, as is the case for most covalent semiconductors. This defect tolerance is a major contributor to the high performance of perovskite-based optoelectronic devices.^[36]

However, the fact that defects have a relatively shallow energy level means that they have a low defect-formation energy, leading to a high concentration of defects. Defects induce ion migration, leading to the creation of new defects.^[37] Thus, even a small defect can induce the formation of a chain of defects, leading to the accumulation of more defects. In particular, halide vacancies, or interstitials are known to promote halide migration by providing a low-energy migration pathway, thereby inducing halide segregation. In addition, halide interstitial defects induce the formation of I_0 , which collides with another I_0 to form I_2 . This I_2 tends to migrate to the surface or grain boundary. Therefore, many defects are concentrated on the surface.^[38] Thus, shallow traps can also be transformed into deep traps, which explains why defects in perovskite should never be overlooked.

Defects also influence charge transport, which has a significant impact on device performance. Photoexcited carriers can either undergo recombination or participate in charge transport. When charge transport is not smooth, it can lead to recombination, thereby reducing device performance. Free carriers are accelerated by an electric field, and during this time, defects act as scatters, altering the acceleration vector through interactions with free carriers.

Defects also trap carriers, reducing overall conductivity and affecting the formation of a potential barrier. The extent to which

photoexcited carriers are trapped is influenced by the density of photoexcited carriers and the trap density. If there are much more photoexcited carriers than traps, the traps become fully occupied by carriers, and the impact on charge transport becomes almost negligible. On the other hand, if there are many traps, most carriers are trapped and repeat hopping or multiple trapping and release, and this process is activated by thermal energy.^[39]

To mitigate these effects, defect passivation strategies are employed. Passivation involves the introduction of specific chemicals or treatments that suppress electrical and chemical activities of defects, allowing for better charge carrier transport and improving the overall performance and stability of the solar cell (Figure 4c).

4. Degradation of Perovskite Crystals

Perovskite can easily be degraded by various internal or external stimuli, resulting in various defects (Figure 5a). Here, we explore the various causes of perovskite degradation and the resulting defects.

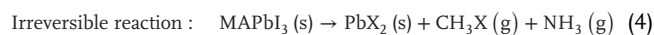
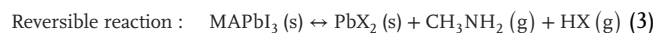
4.1. Light

Under illumination, perovskites can undergo halide segregation and cation segregation. Hoke et al. demonstrated that perovskites with the formula $APb(Br_xI_{1-x})_3$ can separate into a Br-rich phase and an I-rich phase when illuminated.^[40] However, recent high-efficiency PSCs employ $FAPbI_3$, which does not incorporate Br^- ions as halide constituents, eliminating concerns regarding halide segregation. Additionally, halide segregation is a reversible process and does not result in long-term degradation.

Although halide segregation may not be a significant issue for certain perovskite compositions, light can still have other effects on perovskites. Some studies have shown that illumination can increase the concentration of halide vacancies. Kim et al. observed that under illumination, electron-hole separation occurs and the partial equilibrium reaction of holes with the iodide sublattice creates an effective chemical potential for iodine removal, leading to decomposition.^[41] Moreover, light can promote the oxidation of perovskite crystals. With both light and oxygen, electrons, and holes are photogenerated within the perovskite.^[42] These electrons can be extracted by O_2 molecules, forming superoxide (O_2^-) (Figure 5b). Subsequently, the superoxide species can deprotonate the perovskite cations, resulting in the formation of PbI_2 . The formation of these superoxides in iodine vacancies is favored by their low formation energy.^[43] Therefore, the passivation of iodine vacancies is a strategy that can improve the stability and efficiency of PSCs.

4.2. Heat

The thermal degradation of the $MAPbI_3$ perovskite can occur in two pathways:^[26]



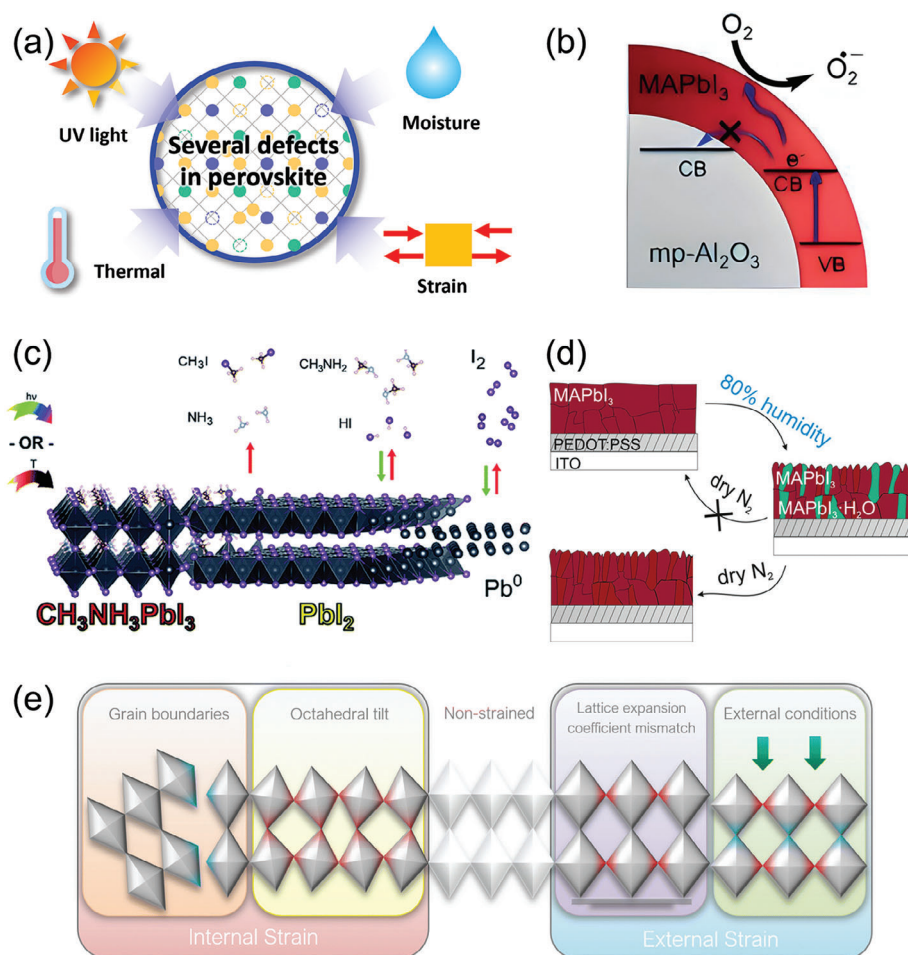


Figure 5. a) Various causes of the perovskite degradation. b) Superoxide formation under illumination. Reproduced with permission.^[42] Copyright 2015, Wiley-VCH. c) Schematic illustrations of the thermal- and light-induced degradations of the perovskite crystal.^[44] d) Schematic illustration of the moisture-induced degradation of perovskite. Reproduced with permission.^[48] Copyright 2016, American Chemical Society. e) Local strain and external strain on the perovskite.^[52]

The first method is an irreversible reaction. However, unless the volatile CH_3NH_2 and HX gases are evaporated and leaked from the encapsulated device (see Figure 5c), the reaction will not lead to permanent degradation, and no chemical equilibrium will be achieved between the formation and destruction of perovskite. Conversely, the second method is irreversible, and the reaction in which CH_3I and NH_3 form MAPbX_3 again is thermodynamically unfavorable;^[44] thus, this reaction can lead to permanent degradation. For MAPbBr_3 , only the second type of degradation has been reported. Conversely, the application of both methods has been reported for MAPbI_3 , and the choice of the degradation method is predominantly determined by various factors.^[45] Therefore, MAPbI_3 may undergo permanent degradation even after encapsulation.

4.3. Moisture

Due to the permeability of H_2O during various stages of perovskite processing, such as material storage, precursor handling, spin coating, and annealing, the impact of moisture on per-

ovskite requires careful investigation. Notably, alkylammonium, and lead salts, which are commonly used in perovskite materials, exhibit hygroscopic behavior, and the hydration water can affect the perovskite film during the deposition process.^[46]

Research has shown that the introduction of a small amount of moisture during the preparation of the perovskite precursor, spin-coating process, or annealing process can improve the film quality. These improvements can be attributed to the enhancement in the perovskite's crystallinity and the increased grain size.

However, when unpackaged PSCs are exposed to air with a relative humidity (RH) higher than 50%, degradation occurs within a few hundred hours. Moisture easily infiltrates the perovskite structure, leading to the formation of intermediate monohydrate and dihydrate perovskite phases. Water molecules within the perovskite crystals form strong hydrogen bonds with organic cations, weakening the bond between the cation and PbI_6 . This increases the deprotonation rate of the organic cation and renders the perovskite more susceptible to degradation from external stimuli, such as heat, or electric fields. Additionally, moisture can protonate iodide, resulting in the generation of volatile hydroiodic acid.^[46]

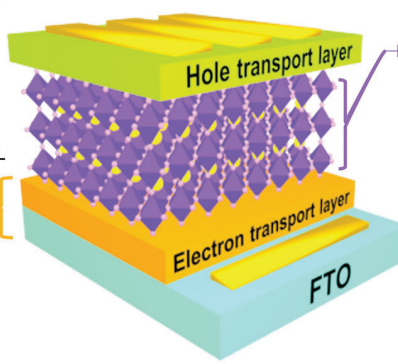
Top layer passivation of perovskite

Problems

- ✓ Surface dangling bonds
- ✓ Electromigration of I^-
- ✓ Decomposition by dopants of the HTL

Requirements of the passivating materials

- ✓ Solubility in orthogonal solvents of the perovskite
- ✓ Appropriate energy level
- ✓ Hydrophobicity



Grain boundary passivation

Problems

- ✓ Dangling bonds in grain boundaries
- ✓ Ion migration through grain boundaries

Requirements of the passivating materials

- ✓ Optimal solubility according to perovskite deposition method

Bottom layer passivation of perovskite

Problems

- ✓ Surface dangling bonds
- ✓ Acceleration of oxidation by metal oxide
- ✓ Lattice mismatch

Requirements of the passivating materials

- ✓ Not dissolved in perovskite solvent
- ✓ Transmission of light in a specific wavelength range
- ✓ Appropriate energy level
- ✓ Appropriate surface energy

Figure 6. Schematic illustration of the PSCs. Problems associated with and requirements of the passivating materials based on the location of the PSCs.

Furthermore, moisture induces the structural decomposition of perovskite crystals.^[47] Li et al. reported that moisture causes the grains of perovskite to disintegrate and form additional grain boundaries (Figure 5d).^[48] Therefore, strategies for understanding, and controlling the effects of moisture on perovskite crystals are required.

4.4. Strain

Perovskite has a mechanically soft nature and is significantly affected by strain. Tensile strain reduces the perovskite defect formation energy as well as the activation energy of ion migration.^[49] That is, when tensile strain is applied to perovskite, defects are more likely to be generated, leading to a decrease in device efficiency. Contrarily, compressive strain enhances the perovskite defect formation energy and enhances the activation energy of ion migration. This suggests that compressive strain can be used to physically control the defects in a device.

In addition to affecting the defect formation, the strain also affects the phase stability. When tensile strain is applied to perovskite crystals, it can induce a transition from the α phase of perovskite to the δ phase.^[50] Therefore, strain control is important in improving the efficiency and stability of PSCs.

Strain in PSCs can be classified into two types: (1) internal strain^[51] and (2) external strain (Figure 5e).^[52] Internally induced strain is caused by a mismatch in the size of the ions in perovskite.^[54] In mixed-halide perovskites, internal strain can also be generated by a mismatch in the nonuniform regions. Meanwhile, externally induced strain is caused by lattice mismatch or thermal coefficient mismatch between layers.^[53] Therefore, special care is required to address each of these causes to improve the performance of PSCs.

5. Passivation Strategy

Generally, passivation is achieved through chemical interaction facilitated by the ionic nature of perovskite. Most defects are positive or negative, and passivation is achieved by the formation of covalent or ionic bonds with the defects to suppress their electrical and chemical activities.

The defects in PSCs have various origins and characteristics, depending on their location within the device. Therefore, it is necessary to employ tailored passivation strategies to address these defects. This review examines location-specific defects and their underlying causes in n-i-p structured PSCs. It also explores the material requirements for each passivation method, as well as the overall passivation method for mitigating these defects effectively (Figure 6).

In this paper, we exclusively focus on n-i-p structured PSCs. Here, we define top-layer passivation as the mitigation of defects within the upper perovskite layer, situated between the HTL and the perovskite itself. Similarly, bottom-layer passivation denotes the mitigation of defects between the ETL and the perovskite, while grain boundary passivation specifically addresses defects existing between crystalline structures within the bulk perovskite.

We begin by discussing the defects in the top layer of the perovskite crystal. The perovskite surface is known to possess numerous dangling bonds that act as traps.^[55] Additionally, defects can accumulate on the top surface of perovskite through ion migration or electromigration processes. Pintilie's group demonstrated that the electric field between the electrode induces the movement of iodine atoms within the perovskite, resulting in the electromigration of I^- ions toward the anode.^[56] Furthermore, Angelis's group employed the density functional theory method and discovered that ion migration primarily occurs on the surface and grain boundaries of perovskite. They also found

that surface passivation can effectively prevent perovskite decomposition by inhibiting the formation of defects induced by ion migration.^[57]

Contrarily, the HTM employed in high-efficiency devices, such as spiro-OMeTAD, requires dopants to achieve high mobility. However, these dopants are hygroscopic and mobile; as such, they adversely affect the perovskite layer. Therefore, it is crucial to develop a passivation method that protects the perovskite from these dopants while inhibiting defect generation.^[58]

Similar to the top layer, the bottom layer of perovskite also contains defects due to dangling bonds. Moreover, in n-i-p PSCs, metal oxide layers, such as TiO₂,^[59] SnO₂,^[60] and ZnO,^[61] are commonly used as ETLs. However, these metal oxides can act as catalysts for perovskite oxidation, leading to degradation.^[62] Consequently, additional defects are generated at the interface between the metal oxide ETL and the perovskite. Developing a passivation method is crucial to address this challenge.

In addition, the charge-extraction rate from the perovskite to the ETL is lower than that from the perovskite to the HTL.^[63] This leads to charge-transport imbalance and charge accumulation, a phenomenon known as the bottleneck effect. To improve the PCE of PSCs, it is important to improve the charge-transport ability at the ETL/perovskite interface. Defects at the interface reduce the charge transport rate and can act as nonradiative recombination centers. Therefore, it is necessary to passivate the interface to improve the charge-transport-layer capability.

Lastly, in bulk perovskite, defects mainly occur at the grain boundaries. Grain boundaries exhibit various dangling bonds, along with the accumulation of deep traps. Park et al. conducted first principles calculations and discovered that split interstitials (I_i⁻), I-I-I trimers, or Pb-I bonds were formed at the grain boundaries, resulting in nonradiative recombination.^[64]

Given the different causes of defects at each location within PSCs, specific strategies are required. This paper aims to provide detailed insights into the passivation strategies for each location.

5.1. Top-Layer Passivation

The passivation materials used on the top layer must meet certain requirements. First, they should be soluble in orthogonal solvents that do not dissolve the perovskite, such as chlorobenzene, or chloroform. This is because the as chlorobenzene, or chloroform. This is because the passivation process occurs after the formation of the perovskite film.

Second, the passivation material needs to have an appropriate energy level for the main charge carriers. However, studies have shown that when the passivation layer is thin, carriers can be transported and extracted effectively through the tunneling effect, even if the energy levels are not perfectly matched. In fact, the presence of a passivation layer can reduce trap-related limitations and increase the extraction efficiency. Additionally, the passivation layer can induce band bending in the perovskite layer, enabling more effective charge extraction.

Last, perovskite is sensitive to moisture; thus, the passivation material should possess hydrophobic properties to prevent moisture ingress from external sources or hygroscopic dopants in the HTL. However, Kim et al. discovered that hydrophilic passivation materials can also be effective in protect-

ing the vulnerable perovskite film. These materials can act as sacrificial layers that absorb moisture instead of the perovskite itself.^[63]

Various materials for passivating the top surface of perovskite have been proposed (Table 1). Snaith and coworkers were the first to report perovskite-surface passivation methods. They induced supramolecular halogen bonding by introducing IPFB to the perovskite surface (Figure 7a). The device with IPFB exhibited an improved PCE of 15.7% with a significant increase in FF (57% to 67%).^[22] The improvement in FF was achieved because IPFB prevented charge accumulation by passivating through electrostatic attraction with the uncoordinated halide ions that act as perovskite traps.

Most typically, Lewis acids, or bases can passivate perovskite. Since perovskite defects are electron-deficient or electron-rich, they can be passivated by forming a coordinate bond between them and a Lewis acid or base. Fullerene and its derivatives are typical Lewis acids that passivate perovskite defects, such as PbI₃⁻, negative-charged Pb-I antisite defects, or undercoordinated halide ions. The spherical shape of the material allows it to accept an external electron efficiently. Huang and coworkers introduced [6,6]-Phenyl-C61-butyric acid methyl ester (PCBM) to passivate the dangling bonds and suppress charge recombination at the material surface.^[65] The film (ITO/PEDOT:PSS/perovskite/(PCBM)) was subjected to photoluminescence (PL) measurement to identify the trap-passivation performance. When the PL laser was directed toward the ITO, the wavelengths of the perovskite film without PCBM and the treated film were the same at 782 nm. Conversely, when the PL laser was directed toward the air side, the wavelength of the perovskite blue-shifted after treatment with PCBM (782 to 775 nm). This indicates that the perovskite film was effectively trap-passivated by PCBM. Park et al. proposed a surface passivation method involving the use of 2,3,5,6-tetrafluoro-7,7,8,8-tetracyanoquinodimethane (F4TCNQ), which is known as a Lewis acid. F4TCNQ not only passivated the defects on the perovskite surface, but also achieved interfacial p-type doping, forming an interfacial dipole layer (Figure 7b-d).^[66] Consequently, the recombination on the surface was effectively limited, and the PCE of the device improved from 14.3% to 16.4%. In addition, because the long-term stability test was conducted without encapsulation in ambient air, the PSCs treated with F4TCNQ exhibited improved stability.

Conversely, Snaith, and coworkers employed thiophene and pyridine, which are Lewis bases, to passivate the perovskite surface (Figure 7e).^[67] These Lewis bases interacted with the undercoordinated Pb ions in the perovskite surface. The passivated device exhibited reduced nonradiative decay, and the PCE of the device increased from 13% to 15.3% and 16.5%. Yang et al. attempted to passivate defects that were particularly abundant on the surface of perovskite using a Lewis base. Through the calculation of the defect formation energy, Yang et al. discovered that the formation energy of the Pb_i antisite defect on the perovskite surface is lower than that of the bulk Pb_i antisite defect. To address this issue, they attempted to passivate the Pb_i antisite defects on the perovskite surface using theophylline, which contains a Lewis base C=O group. The theophylline molecule also featured an imidazole ring that formed hydrogen bonds with iodine atoms, thereby strengthening the adsorption with the Pb_i defect. Due

Table 1. Summary of the performance of the PSCs with top layer passivation.

Passivator	Type	Perovskite		J_{sc} [mA cm ⁻²]	V_{oc} [V]	FF [%]	Efficiency [%]	Reference
Iodopentafluorobenzene	Lewis acid	MAPbX ₃	Control	23.4	1.02	57.0	13.0	[19]
			Target	23.4	1.06	67.0	15.7	
2,3,5,6-tetrafluoro-7,7,8,8-tetracyanoquinodimethane (F4TCNQ)	Lewis acid	MAPbI ₃	Control	19.4	1.04	69.9	14.3	[61]
			Target	20.3	1.06	75.4	16.4	
Pyridine	Lewis base	MAPbI ₃	Control	20.7	0.95	68.0	13.1	[62]
			Target	24.1	1.05	72.0	16.5	
Theophylline	Lewis base	(FAPbI ₃) _{0.92} (MAPbBr ₃) _{0.08}	Control	24.8	1.17	72.9	21.0	[63]
			Target	25.2	1.19	78.1	23.5	
Quaternary ammonium halides	ionic compound	MAPbI _{3-x} Cl _x	Control	22.5	1.05	73.0	17.1	[65]
			Target	22.7	1.08	80.0	19.6	
PEAI	ionic compound	FAPbI ₃	Control	24.3	1.12	79.1	21.5	[68]
			Target	25.2	1.18	78.4	23.3	
Dodecylammonium formate	ionic compound	FAPbI ₃	Control	25.0	1.09	76.6	20.8	[7a]
			Target	25.3	1.20	82.2	25.0	
PEA ₂ PbI ₄	2D perovskite	Cs _{0.05} (MA _{0.13} FA _{0.87}) _{0.95} Pb(I _{0.87} Br _{0.13}) ₃	Control	22.8	1.05	71.0	17.0	[67]
			Target	22.9	1.11	73.0	18.5	
Methylthiotriphenylamine-substituted copper phthalocyanine (SMe-TPA-CuPc)	HTM	Cs _{0.05} (MA _{0.13} FA _{0.87}) _{0.95} Pb(I _{0.87} Br _{0.13}) ₃	Control	23.4	1.14	76.9	20.6	[70]
			Target	23.7	1.14	79.3	21.5	
Bithiophene-Imide-Based Polymer (PFBTI)	HTM	Cs _{0.05} FA _{0.95} PbI ₃	Control	24.9	1.10	78.2	21.4	[71]
			Target	24.6	1.16	80.8	23.1	
Al ₂ O ₃	Anti-aging material	Cs _{0.05} (MA _{0.17} FA _{0.83}) _{0.95} Pb(I _{0.83} Br _{0.17}) ₃	Control	23.8	1.14	78.3	21.3	[7e]
			Target	22.7	1.12	87.8	21.2	

to this passivation strategy, the PCE of the PSC increased from 21.02% in the control device to 22.32% in the theophylline-treated device.^[68]

Ionic compounds have also been employed for passivation through their interaction with defects in perovskite.^[69] Huang's group passivated defects in perovskite using quaternary ammonium halide anions (QAHs).^[70] The calculated density of states showed that QAHs can passivate both types of charged ionic defects (a cationic Pb cluster on a MAI-terminated surface and an anionic Pb-I antisite defect on a PbI₂-terminated surface). Resultantly, the PSCs treated with QAHs exhibited a significant increase in efficiency from 19.2% to 21.0% compared with the previously reported PCBM-passivated PSCs. Additionally, in particular, the V_{oc} increased significantly from 1.03 to 1.14 V. Kim et al. explored interfacial engineering using alkylammonium formates (AAFs) of different lengths consisting of large organic cations and pseudo-halide anions (Figure 7f,g).^[10] AAFs undergo a liquid phase transition at the annealing temperature of perovskite, and this liquid state induces selective and effective passivation. PSCs using dodecylammonium formate exhibited a significantly higher PCE of 25.0% compared with the control device (PCE = 20.76%).

The formation of a 2D perovskite surface layer has also been suggested as a method of passivation.^[71] In addition, since the 2D perovskite passivation layer is hydrophobic and has a long cation part, it can prevent moisture from penetrating the vulnerable 3D perovskite. It can also effectively prevent ion migration,

improving stability. Chen et al. proposed a 2D PEA₂PbI₄ capping layer to passivate the top surface of perovskite (Figure 8a).^[72] The 2D PEA₂PbI₄ layer improved the V_{oc} of PSCs from 1.05 to 1.11 V by enhancing the QFLS and inhibiting nonradiative recombination. In addition, the 2D PEA₂PbI₄ layer improved stability by suppressing water permeation and ion migration. Thus, the PSCs treated with the 2D PEA₂PbI₄ layer under ambient conditions with a RH of 60 ± 10% maintained close to 90% of their initial efficiency even after 1000 h of operation (Figure 8b). However, it has been reported that passivation using organic ammonium halide itself, a material for the 2D perovskite passivation layer, is more effective than forming a 2D perovskite. You's group compared the passivation methods using PEA₂PbI₄ and the 2D PEA₂PbI₄ layer.^[73] After the perovskite was treated with PEA₂PbI₄, XRD analysis confirmed the formation of PEA₂PbI₄ in the film heated to 100 °C, but not in the unheated film. PL and time-resolved PL measurements revealed that when PEA₂PbI₄ itself was used, a comparatively large PL intensity and long lifetime were achieved, indicating that it is a more efficient passivation material.

A method of passivating the surface of perovskite using a dopant-free HTM with a functional group capable of defect passivation has been reported.^[74] This method does not require an additional process to form a passivation layer. Qu et al. introduced methylthiotriphenylamine-substituted copper phthalocyanine (SMe-TPA-CuPc) as a dopant-free HTM (Figure 8c).^[75] The sulfur atoms in SMe-TPA-CuPc effectively passivated the

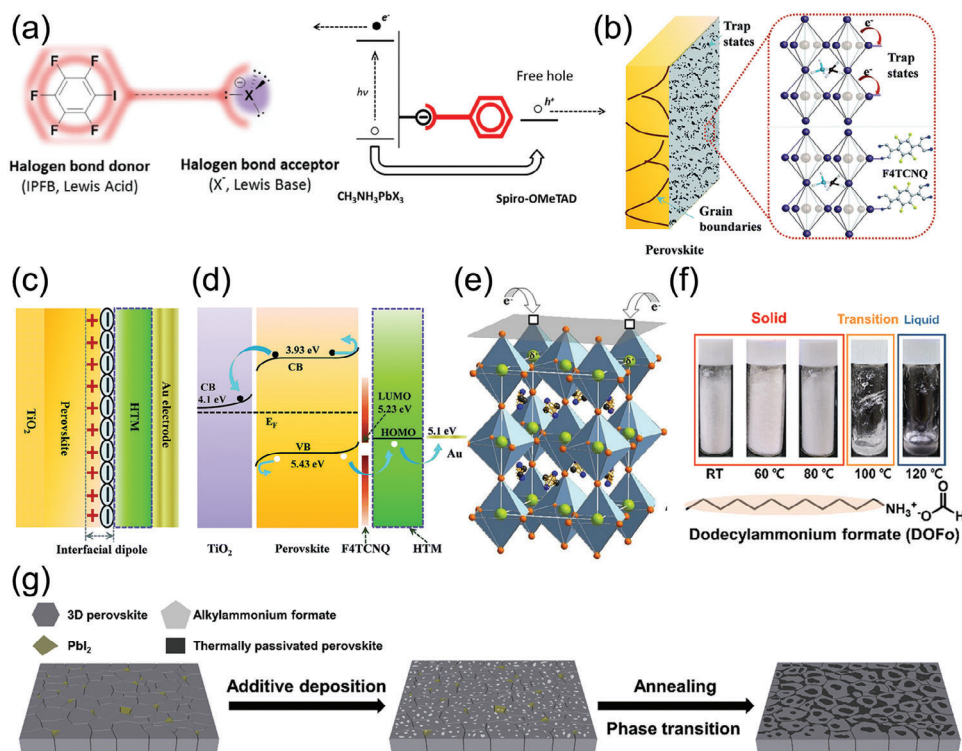


Figure 7. a) Halogen bonding passivation. Reproduced with permission.^[22] Copyright 2014, American Chemical Society. b) Schematic illustration of the trap passivation using F4TCNQ. c) Schematic illustration of interfacial dipole formation due to F4TCNQ. d) Schematic illustration of the band bending due to F4TCNQ.^[66] e) Schematic illustration of the defect passivation mechanism of the perovskite crystal with Lewis bases. Reproduced with permission.^[67] Copyright 2014, American Chemical Society. f) Schematic image of the phase transition of dodecyl ammonium formate according to temperature. g) Schematic illustrations of the phase transition-induced passivation of the perovskite film.^[10]

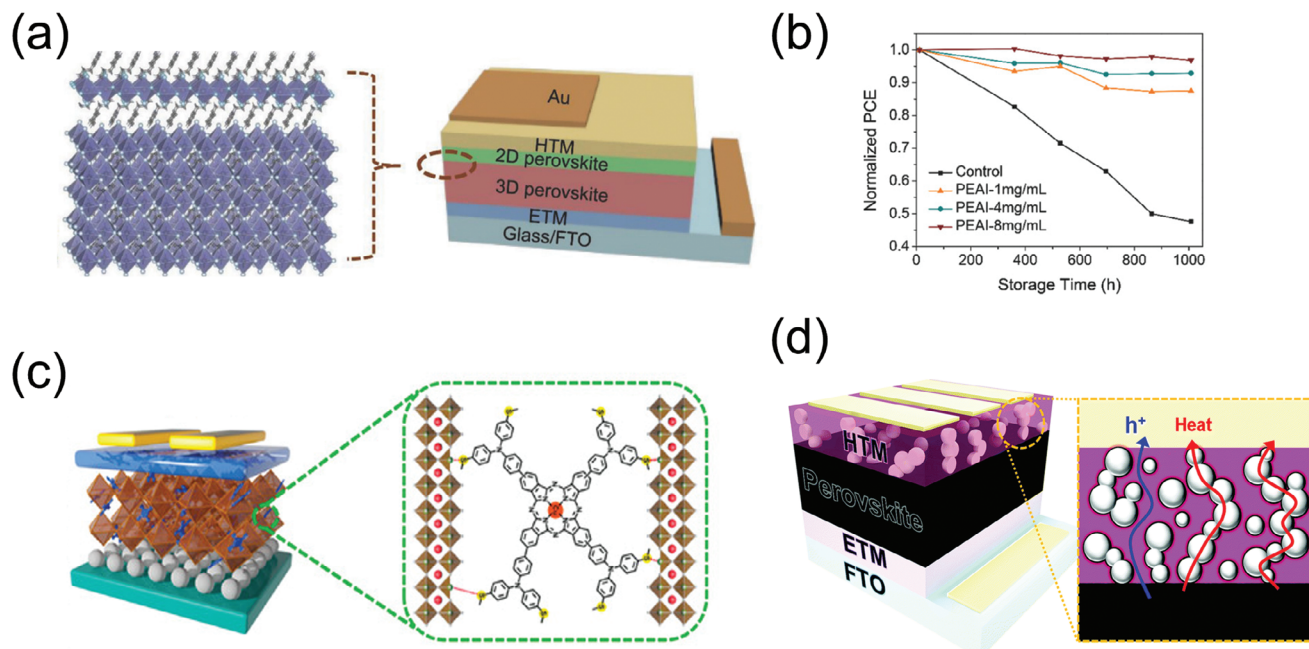


Figure 8. a) Schematic illustrations of the 2D perovskite passivation. b) Stability test of the PSCs with the 2D PEA₂PbI₄ layer under ambient conditions, with RH of 60% ± 10%. Reprinted with permission.^[72] Copyright 2014, Wiley-VCH. c) Schematic illustration of the methylthiotriphenylamine-substituted copper phthalocyanine (SMe-TPA-CuPc)-based perovskite solar cells. Reproduced with permission.^[75] Copyright 2014, Wiley-VCH. d) Schematic illustration of the PSCs with Al₂O₃ NPs.

perovskite because sulfur is a soft base that interacts strongly with the soft acid, Pb_2^+ , according to the soft–hard theory. Consequently, the device exhibited a PCE of 21.51%, which is higher than that of the reference, i.e., the doped spiro-OMeTAD-based device (20.61%). Bai et al. applied dopant-free bithiophene-imide-based polymeric HTMs (PFBTI) in PSCs. PFBTI has a $\text{C}=\text{O}$ group and a fluorine atom that can act as a Lewis base. These effectively passivated the undercoordinated Pb_2^+ . Thus, the PSCs with PFBTI exhibited a high PCE of 22.2%, even without a dopant.^[76]

A method of limiting defect generation by reducing aging stress rather than chemical interaction was also proposed. Choi et al. attempted to effectively dissipate the heat accumulated in a PSC by introducing an Al_2O_3 nanoparticle with a high thermal conductivity (Figure 8d). The PSCs incorporating the Al_2O_3 nanoparticles exhibited significantly reduced trap-assisted recombination.^[10]

5.2. Bottom-Layer Passivation

The materials employed for bottom-layer passivation in PSCs also typically need to meet several conditions. First, the passivation material should not dissolve in the solvent of the perovskite, such as dimethylformamide (DMF) or dimethyl sulfoxide (DMSO). Since the perovskite layer is deposited after the passivation layer, it is crucial that the passivation material remains intact and does not dissolve during the perovskite deposition process. This is important because if the passivation material dissolves in the perovskite solvent, it will be difficult to deposit the passivation layer uniformly. The uniform deposition of the passivation layer is essential for avoiding defects in the perovskite layer and maintaining a high PCE.

Second, the passivation layer at the bottom should allow for the transmission of light in a specific wavelength range. Since light reaches the bottom-layer passivation material first, light transmission capability is also required. If the passivation material absorbs a significant amount of the light meant for the perovskite, it can reduce the PCE of the solar cell. Therefore, the bottom passivation material should exhibit minimal absorption in the wavelength range of 300–800 nm, where the perovskite absorbs a significant portion of light. However, if the passivation layer is very thin, even if the passivation material absorbs some light in the perovskite absorption range, the impact on the PCE might be insignificant. This allows for its use with the restrictions on thickness.

Thirdly, similar to the top-layer passivation material, the bottom-layer passivation material should possess an appropriate energy level for efficient charge transfer. However, for a thin bottom passivation layer, charge transport can still occur effectively through the tunneling effect, even if the energy levels are not perfectly matched. However, this should be used with caution because even a slight increase in thickness can drastically decrease the efficiency.

Lastly, the surface energy of the bottom passivation layer is crucial as it can influence the crystallization of the perovskite layer. If the bottom layer is overly hydrophilic, the perovskite crystals formed tend to be small, resulting in an increased number of grain boundaries, which can adversely affect efficiency. Con-

versely, if the bottom layer is excessively hydrophobic, it can lead to the formation of voids, defects, and an inhomogeneous perovskite film. Therefore, achieving an appropriate surface energy is vital for obtaining high-quality perovskite films and maximizing the PCE.

Various materials and strategies have been explored for the bottom-layer passivation of PSCs (Table 2). Lewis acid–base interactions have also been explored for the bottom-layer passivation. Yang et al. applied a fullerene derivative (PCBB-2CN-2C8) as an interlayer for bottom-layer passivation (Figure 9a).^[77] They used C_{60} with high electron affinity as the main building block for efficient electron transport, attached electron-deficient cyano groups to passivate oxygen vacancies, and employed long dioctyloxy chains to avoid melting in the polar perovskite precursor solvent. Consequently, the PSCs treated with PCBB-2CN-2C8 exhibited a higher PCE (17.35%) than the control device (14.38%). The study demonstrated a rational molecular design strategy for bottom passivation.

Ionic compounds have also been employed for the bottom-layer passivation of perovskite crystals.^[78] Li et al. introduced CsBr to reduce direct surface contact between the perovskite and TiO_2 , thereby inhibiting TiO_2 from playing a catalytic role in the degradation of perovskite (Figure 9b,c).^[79] This considerably increased the ultraviolet (UV) stability of the PSCs. Wu et al., in a recent study, employed P-biguanylbenzoic acid hydrochloride (PBGH) as a modification agent for the SnO_2 /perovskite interface in PSCs. The addition of PBGH improved the film quality of the perovskite layer by passivating the dangling bonds of Sn and interacting with PbI_2 . Resultantly, the PSC treated with PBGH exhibited a significantly higher PCE (24.79%) compared with the control device (22.58%). These findings highlight the effectiveness of PBGH in enhancing the performance of PSCs through interface modification.^[78]

A method for passivation through modified metal oxide ETLs has also been proposed. Wu et al. proposed a strategy for passivating perovskite by capping TiO_2 with chlorine (Cl-TiO_2).^[80] When Cl treatment was applied, the Pb–I antisite defect was localized near the valence band edge. Conversely, the Pb–Cl antisite defect has a relatively high formation energy, is shallow, and is delocalized. Therefore, interfacial recombination was suppressed when Cl-TiO_2 was used, and the PCE of the device increased from 18.5% to 20.9%. Seok and coworkers introduced a Cl-bonded SnO_2 (Cl-bSO) layer to Cl-containing FAPbI_3 PSCs.^[81] Between the Cl-bSO layer and the Cl-containing FAPbI_3 , the FASnCl_x layer, which is the coherent interlayer, was formed. This coherent interlayer inhibited the nonradiative recombination and improved the PCE of the PSCs to 25.8%.

A self-assembled monolayer (SAM) is an effective tool for passivating the bottom layer of perovskite.^[82] The SAM refers to a molecular assembly spontaneously generated on a substrate by adsorption. SAM molecules are composed of an anchoring group, a linker, and a functional group. In n–i–p-structured PSCs, the anchoring group is adsorbed on the oxide transport layer, and the functional group undergoes an interaction in the perovskite to effectively passivate the traps present at the interface of the perovskite. Snaith and coworkers introduced a fullerene self-assembled monolayer (C_{60}SAM) onto a mesoporous TiO_2 layer (Figure 9d).^[83] C_{60}SAM prevented the back transport of electrons from TiO_2 to the perovskite through

Table 2. Summary of the performance of the PSCs with bottom layer passivation.

Passivator	Type	Perovskite		J_{SC} [mA cm ⁻²]	V_{OC} [V]	FF [%]	Efficiency [%]	Reference
a triblock fullerene derivative (PCBB-2CN-2C8)	Lewis acid-base	MAPbI ₃	Control	20.2	0.99	72.2	14.4	[72]
			Target	20.7	1.06	79.1	17.4	
CsBr	Ionic compound	MAPbI _{3-x} Cl _x	Control	18.7	0.99	69.0	13.1	[74]
			Target	20.7	1.06	75.0	16.3	
P-biguanylbenzoic acid hydrochloride (PBGH)	Ionic compound	FA _{0.9} Cs _{0.1} PbI ₃	Control	24.9	1.17	78.7	22.9	[73b]
			Target	25.0	1.20	82.8	24.8	
TiO ₂ -Cl	modified metal oxide ETL	FA _{0.85} MA _{0.15} PbI _{2.55} Br _{0.45}	Control	21.6	1.13	0.76	18.5	[75]
			Target	22.3	1.19	80.6	21.4	
FASnCl _x	modified metal oxide ETL	FAPbI ₃ :0.38MDACL ₂	Target*	25.7	1.19	84.4	25.8	[76]
Fullerene self-assembled monolayer (C ₆₀ -SAM)	SAM	MAPbI _{3-x} Cl _x	Control	19.4	0.82	65.0	10.2	[78]
			Target	19.6	0.84	72.0	11.7	
3-aminopropanoic acid SAM	SAM	MAPbI ₃	Control	19.1	0.99	63.0	12.0	[79]
			Target	22.5	1.07	65.0	15.7	
3-(1-pyridinio)-1-propanesulfonate	Zwitterion	Cs _{0.05} (MA _{0.17} FA _{0.83}) _{0.95} Pb(I _{0.83} Br _{0.17}) ₃	Control	23.0	1.10	77.6	19.6	[20]
			Target	23.6	1.16	78.4	21.4	
zeolitic imidazolate framework-8 (ZIF-8)	Control crystallization	FAPbI ₃	Control	23.8	1.16	78.6	21.6	[81]
			Target	25.6	1.18	80.1	24.0	

energy-level alignment rather than by directly controlling the defects through interactions in perovskite (Figure 9e). Zuo et al. introduced 3-aminopropanoic acid SAM (C₃-SAM) onto ZnO layers^[84] (Figure 9f). C₃-SAM formed a permanent dipole moment, inducing band bending, which facilitated charge extraction, and improved the quality of the perovskite crystal film. Scanning electron microscopy images show that the number of pinholes in the perovskite film treated with C₃-SAM is significantly lower than that in the untreated film (Figure 9g). Consequently, PSCs treated with C₃-SAM exhibit improved performance.

A zwitterion is a molecule or material comprising immobilized cationic and anionic charges; the part with an anionic charge binds to an oxide-based ETL, whereas the part with a cationic charge interacts with a bond present on the perovskite surface to form a SAM. The 3-(1-pyridinio)-1-propanesulfonate zwitterion was introduced onto the SnO₂ layer of PSCs by Park's group (Figure 9h). The zwitterion passivated the defects in the perovskite crystal film, suppressed charge recombination, improved charge extraction, and increased the device's thermal stability.^[23] The PSCs subjected to zwitterion treatment exhibited a PCE of 21.43% and showed excellent thermal stability (Figure 9i,j).

Bottom-layer passivation also reduces the strain induced by the difference in the lattice size and thermal expansion coefficient between the inorganic ETL and the perovskite. The tensile strain generated in the interlayer reduces the defect formation energy of perovskite and the activation energy of ion migration. Several researchers have devoted intensive efforts to solving this problem. Wu et al. incorporated polystyrene (PS) in PSCs to release the residual stress in the perovskite/ETL interface (Figure 10a).^[85] As PS has a low T_g, it can reduce the stress during the annealing process. In addition, it can passivate the defects of the SnO₂ surface and improve stability by forming an inner-encapsulated perovskite layer.

A defect engineering method that involves tailoring the crystallization of perovskite has also been reported. Sheng et al. introduced a SnO₂/PbI₂ layer (the first step in the two-step perovskite deposition process) to the zeolitic imidazolate framework-8 (ZIF-8@FAI) surrounding the FAI (Figure 10b). The perovskite film incorporated with ZIF-8@FAI exhibited a higher porosity than the control film, which benefited the subsequent intercalation reaction. ZIF-8@FAI not only reduced the residual strain, but also effectively passivated uncoordinated Pb₂⁺ ions, leading to improvements in both PCE and stability of the PSCs.^[86]

5.3. Bulk Perovskite Passivation

There are two methods for perovskite deposition, and the optimal bulk perovskite passivation strategy depends on the method employed. In the one-step spin-coating method, two types of solvents are employed to deposit perovskite. One is a precursor solvent that dissolves the perovskite material, such as DMF, or DMSO. The other is an antisolvent that controls the crystallinity of perovskite, such as diethyl ether, or chlorobenzene. Therefore, passivation materials can be applied in two ways: adding them to the precursor solvent or adding them to the antisolvent. When a passivation material is added as a precursor additive, it is evenly distributed throughout the perovskite crystal, effectively passivating the entire crystal. Conversely, when added as an antisolvent additive, the passivating material mainly locates in the upper part of the perovskite crystal, with minimal presence in the lower part. Long et al. employed a nonfullerene small molecule (F8IC) as a precursor and antisolvent additive and examined its passivation effects on perovskite (Figure 11a).^[87] They found that adding F8IC as an antisolvent additive prevented nonradiative recombination in the upper part of the perovskite, improving efficiency. However, this method was not as effective as adding F8IC

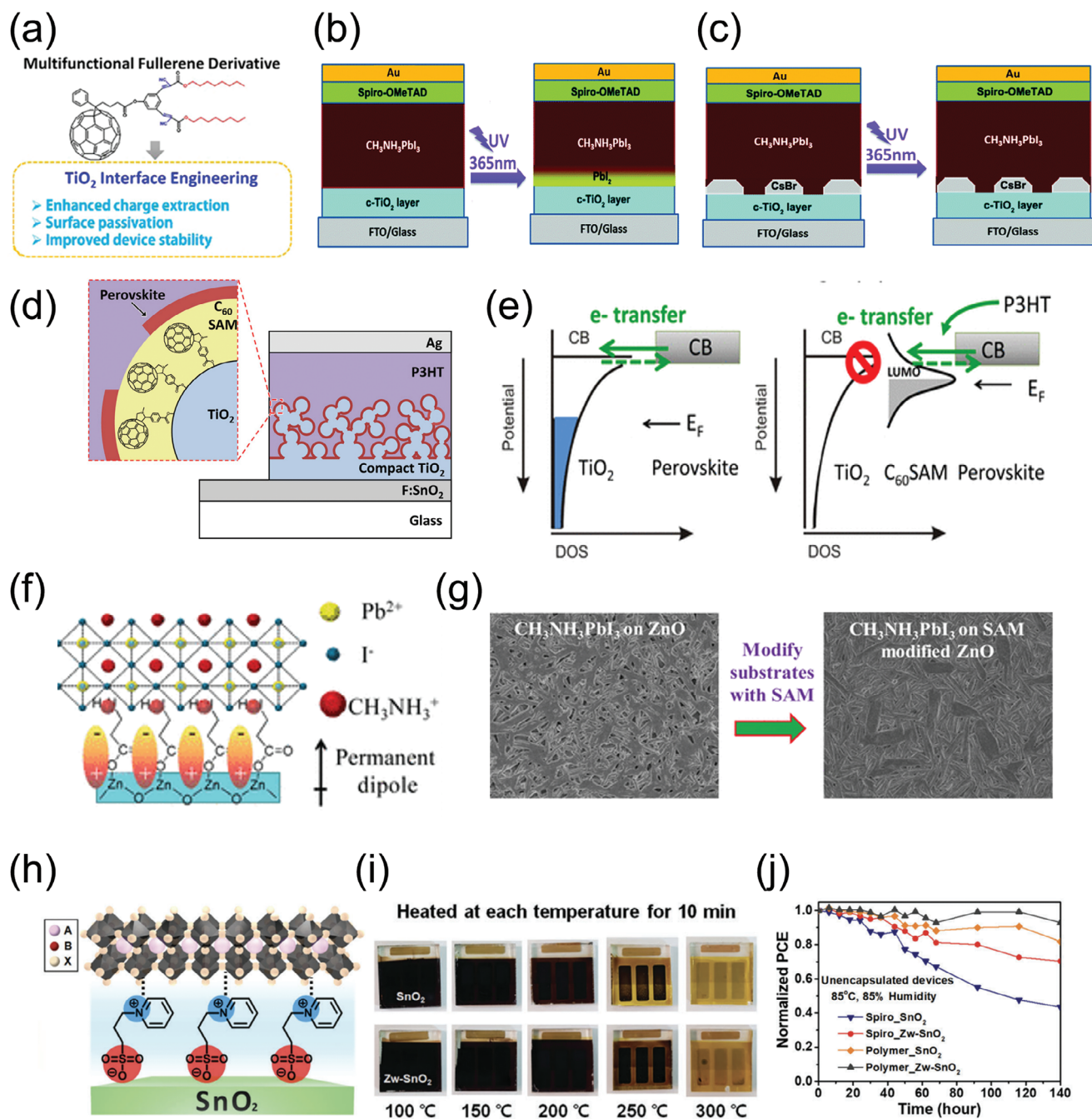


Figure 9. a) Design strategy of the fullerene derivative passivator. Reproduced with permission.^[77] Copyright 2015, American Chemical Society. Comparison of the PSCs b) with and c) without Cs Br modification.^[79] d) Schematic illustration of the PSCs with C₆₀SAM. e) Energy-level diagram of the control device and C₆₀SAM-based device. Reproduced with permission.^[83] Copyright 2015, American Chemical Society. f) Schematic illustration of the perovskite with C₃-SAM. g) SEM image of the perovskite with and without C₃-SAM. Reproduced with permission.^[84] Copyright 2015, American Chemical Society. h) Schematic illustration of the zwitterion-treated SnO₂. i) PSCs with the pristine SnO₂ and Zw-SnO₂ under thermal stress. j) Stability test of the PSCs with the pristine SnO₂ and Zw-SnO₂.^[23]

as a precursor additive, which resulted in the comprehensive passivation of the perovskite defects and the formation of compact and dense perovskite crystals. Notably, not all nonfullerene n-type organic semiconductor passivators yield the same result when added as a precursor. The passivation method must be selected considering the energy level.

In the two-step deposition method, two solvents are also used. One is a precursor that dissolves lead halide (PbX₂), such as DMF, or DMSO. The other is a precursor that dissolves AX (where A is a cation, and X is a halide), such as isopropyl alcohol.

The passivation pattern differs depending on whether the passivator is added to the PbX₂ precursor or the AX precursor. The

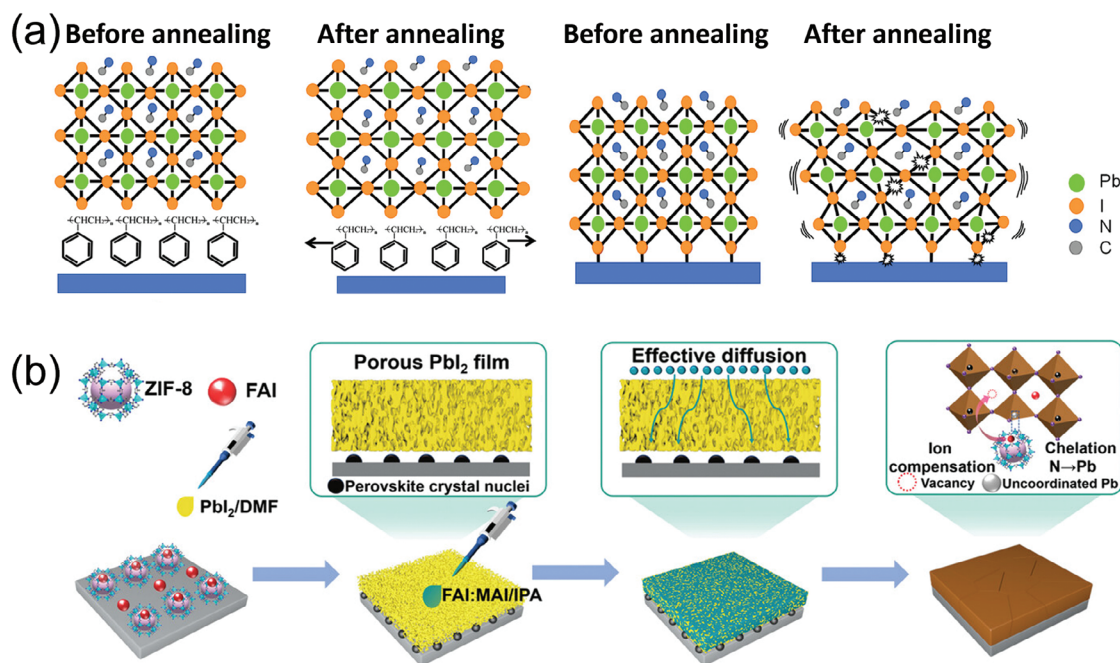


Figure 10. a) Schematic illustration of the perovskite film with and without PS before and after annealing. Reproduced with permission.^[85] Copyright 2019, Wiley-VCH. b) Schematic illustration of the ZIF-8@FAI treatment. Reproduced with permission.^[86] Copyright 2023, Wiley-VCH.

additive dissolved in the PbX_2 precursor interacts with Pb^{2+} , and it helps control the crystallization of perovskite and the formation of a uniform film and large grains. Contrarily, the passivator dissolved in the AX precursor controls the rough PbI_2 film,^[88] suppresses the generation of residual I,^[89] and mainly locates in the grain of the PbI_2 crystal, enabling the selective passivation of only the perovskite grain boundary defect.

In addition, the energy level of the passivation material should be considered. Niu et al. introduced various p-type and n-type semiconducting molecules to passivate bulk perovskite.^[90] When the highest occupied molecular orbital of the p-type semiconducting molecule is energetically close to the perovskite VB maximum and the n-type lowest unoccupied molecular orbital is close to the perovskite CB maximum, the trap state can be made shallow, and the carriers can be easily transported through the semiconducting bridge between the grains. Conversely, if the energy disparity is significant, it acts as an energy barrier, hindering intergrain carrier transport (Figure 11b). Therefore, it is good to consider the energy level during bulk perovskite passivation.

To date, various materials have been introduced for bulk perovskite passivation (Table 3). One common approach is to use Lewis acids or bases, which effectively passivate bulk perovskite defects.^[90,91] For example, Sargent et al. reported a precursor additive method using PCBM. The PCBM passivated the Pb–I antisite defects effectively (Figure 11c).^[92] They measured the secondary ion mass spectrometry depth profile to determine the distribution of PCBM in the device to passivate the defects (Figure 11d). The results showed that PCBM was evenly distributed regardless of the crystal depth (Figure 11e). The PSCs incorporating the PCBM additive exhibited a PCE of 14.4%, higher than that of the control PSC (6.7%). Zhao et al. employed a novel small molecule of indaceno[1,2-b:5,6-b']dithiophene (IDT) with

bilateral arms of oxindole (IDT-OD) as a bulk perovskite passivation material.^[93] The S molecule and carbonyl group of the IDT unit acted as Lewis bases and passivated the undercoordinated Pb^{2+} , whereas the N–H group passivated the halide dangling bond. The IDT-OD improved the PCE of the PSCs from 22.8% to 24.04%.

Another approach involves using polymers as perovskite passivators.^[94] Polymers can confer hydrophobic properties on the perovskite film, preventing moisture penetration, and ion migration. Zuo et al. applied a polymer modifier through the interdiffusion method to enhance the PCE of a PSC.^[95] In their approach, they initially mixed poly(4-vinylpyridine) (PVP) with the perovskite precursor prior to device fabrication. However, they encountered difficulties in passivating PVP by directly mixing it with the perovskite precursor, as the pyridine in PVP strongly interacted with the Pb ions, leading to gelation reactions. To overcome this challenge, they employed the interdiffusion method. The interdiffusion method involved the formation of a mesoporous PbI_2 layer as the initial step. Subsequently, PVP was mixed with a methylammonium iodide (MAI) solution and applied onto the PbI_2 layer (Figure 11f). This approach resulted in a PSC with an improved PCE of 20.2%.

A self-induced passivation method has also been proposed, where passivation is induced through excessive stoichiometry or thermal annealing without the need for additional additives. Yang et al. demonstrated that thermal annealing generates PbI_2 at the perovskite grain boundary, reducing carrier recombination (Figure 12a).^[96] Son et al. achieved self-induced passivation by introducing an additional 6 mol% of MAI, which not only limited nonradiative recombination, but also improved the hole-extraction efficiency via the formation of an ionic conducting pathway at the grain boundary.^[97]

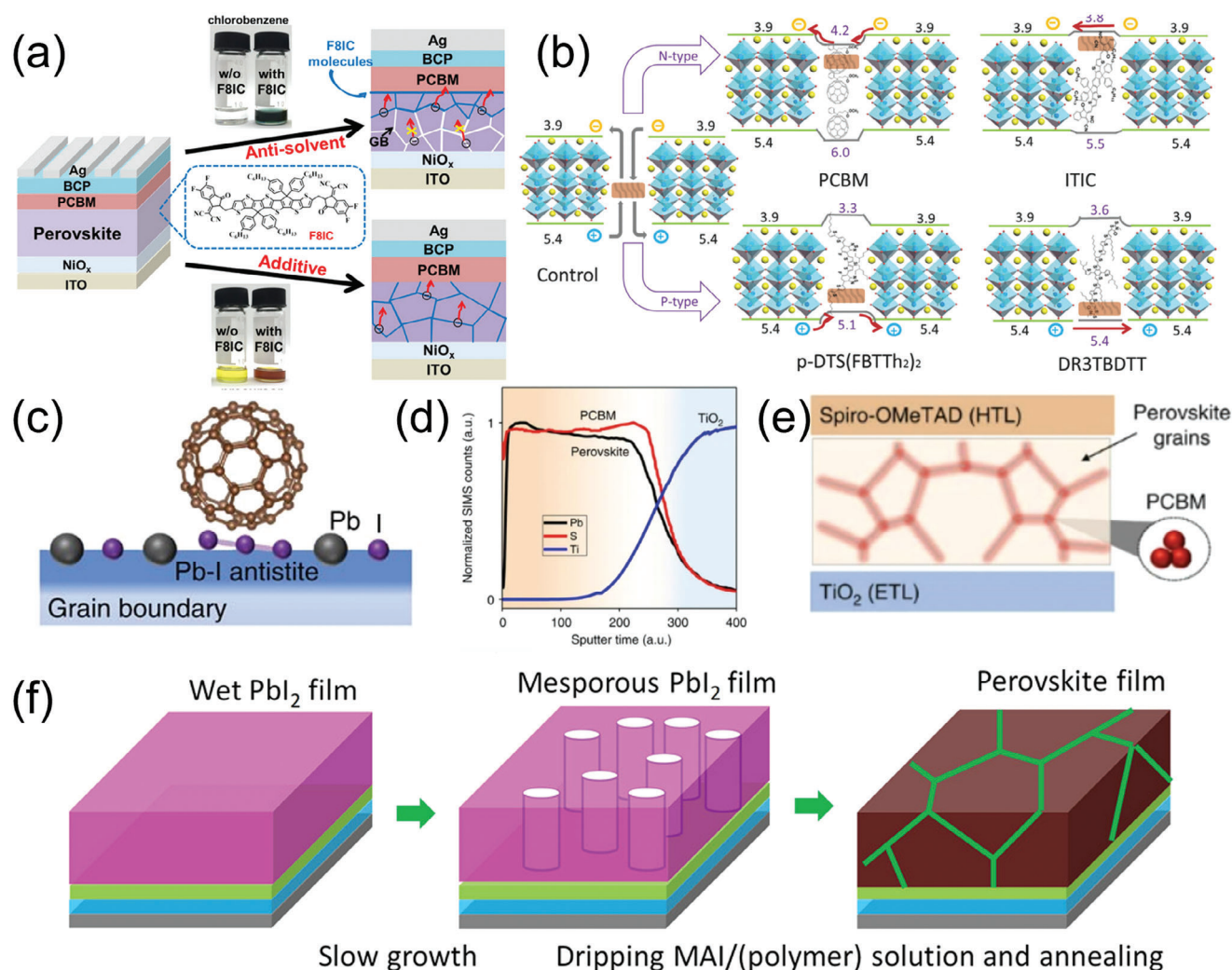


Figure 11. a) Schematic illustration of the F81C passivation using an antisolvent and the additive method. Reproduced with permission.^[87] Copyright 2020, American Chemical Society. b) Schematic illustration of the energy offset of the passivation materials. Reproduced with permission.^[90] Copyright 2018, Wiley-VCH. c) Schematic illustration of the fullerene passivation. d) Secondary ion mass spectrometry (SIMS) depth profile of the PCBM-treated perovskite on the TiO₂ substrate. e) Schematic illustration of the grain boundary passivation by PCBM. f) Schematic illustration of the interdiffusion method of polymer passivation.^[95]

Doping the cation or anion part of perovskite with other ions is another strategy for hindering defect generation. Min et al. attempted to stabilize the α -phase by doping FAPbI₃-based perovskite with MDACl₂.^[98] MDA (methylenediammonium) can successfully stabilize α -FAPbI₃ because it has many H groups and an ionic radius similar to that of FA. FAPbI₃-based PSCs doped with 3.8 mol% of MDACl₂ achieved an efficiency of 23.7%. Kim et al. reduced the strain of perovskite by simultaneously doping it with MDA and Cs cation to suppress the generation of unwanted trap sites.^[99] Jeong et al. carried out anion doping of perovskite using a pseudo-halide material. HCOO⁻ can passivate effectively because its halide vacancy and binding energy are greater than those of the I ion. Consequently, the PSC engineered using pseudo-halides exhibited a high PCE of 25.6%.^[100]

Ionic compounds have been widely employed for bulk perovskite passivation because they not only help passivate defects

but also promote the perovskite film-formation process.^[101] For the bulk perovskite passivation, the ionic compound plays different roles because the cation and anion parts are located in different places during the crystallization process. Liu et al. attempted to passivate PSCs using an amphiphilic π -conjugated ionic compound (QAPyBF₄) (Figure 12b). P-type QAPy⁺ was located all over the perovskite and the passivated defects in the bulk perovskite. Conversely, BF₄⁻ was located at the perovskite/ETL interface, and it induced the desired band alignment (Figure 12c,d). They reported an impressive PCE of 23.1% as a result of their study.^[102]

Two-dimensional perovskite has also been used for bulk passivation. Yang's group introduced 2D PEA₂PbI₄ for the passivation of bulk perovskite.^[103] The 2D PEA₂PbI₄ was spontaneously formed at the grain boundary, and it suppressed moisture penetration and ion migration, limiting defect generation (Figure 12e).

Table 3. Summary of the performance of the PSCs with bottom layer passivation.

Passivator	Type	Perovskite		J_{SC} [mA cm ⁻²]	V_{OC} [V]	FF [%]	Efficiency [%]	Reference
PCBM	Lewis acid-base	MAPbI ₃	Control	13.1	0.97	62.0	6.70	[87]
			Target	18.0	1.09	75.0	14.4	
indaceno[1,2-b:5,6-b']dithiophene with oxindole	Lewis acid-base	FA _{0.85} MA _{0.15} PbI ₃	Control	25.0	1.13	80.5	22.8	[88]
			Target	25.1	1.18	81.4	24.0	
poly(4-vinylpyridine) (PVP)	polymer	MAPbI ₃	Control	21.8	1.09	75.9	18.1	[90]
			Target	21.7	1.15	80.9	20.2	
PbI ₂	self-induced passivation	MAPbI ₃	Control	2.70	0.75	32.2	0.66	[91]
			Target	18.1	0.96	69.3	12.0	
MAI	self-induced passivation	MAPbI ₃	Control	23.8	1.04	71.4	17.7	[92]
			Target	23.6	1.13	77.7	20.7	
MDACl ₂	doping	FAPbI ₃	Control	25.1	1.14	80.6	23.1	[93]
			Target	26.5	1.14	81.8	24.7	
MDA, CsI	doping	FAPbI ₃	Control	26.3	1.14	82.0	24.5	[94]
			Target	26.2	1.17	82.2	25.2	
Formamidine formate	doping	FAPbI ₃	Control	25.7	1.15	80.7	23.9	[95]
			Target	26.4	1.19	81.7	25.6	
Amphiphilic π -conjugated ionic compound (QAPyBF ₄)	ionic compound	(CsI) _{0.05} (FAPbI ₃) _{0.79} (MAPbBr ₃) _{0.16}	Control	23.5	1.16	76.8	20.9	[97]
			Target	23.8	1.20	80.9	23.1	
PEA ₂ PbI	2D perovskite	FA _{0.98} Cs _{0.02} PbI ₃	Control	24.2	1.05	64.6	16.4	[98]
			Target	24.4	1.13	76.5	21.1	

5.4. Double/Triple Passivation

Certainly, defects in perovskite do not occur only at one site; thus, the simultaneous passivation of several sites is beneficial and efficient. Many researchers have proposed studies in which top layer and bottom layer of the perovskite, as well as the bulk perovskite, are simultaneously passivated.^[104]

Peng et al. used a poly(methyl methacrylate (PMMA) film to passivate both ETL/perovskite and perovskite/HTL interfaces. The carbonyl group present in PMMA is a Lewis base, which effectively passivated the uncoordinated Pb²⁺ defect at both interfaces, thereby effectively limiting nonradiative recombination. This approach resulted in a significant improvement in V_{OC} , resulting in a high V_{OC} of 1.22 V.^[105] Mahmud et al. performed BAI-based 2D perovskite passivation on the ETL/perovskite interface and the perovskite/HTL interface. Effective defect passivation was achieved on both interfaces, and interfacial nonradiative recombination was constrained. This 2D passivation layer was not uniformly distributed over all areas and showed incomplete surface coverage, causing charge transport to occur only in the localized unpassivated areas. This incomplete surface coverage makes the 2D passivation strategy effective for both ETL/perovskite and perovskite/HTL interfaces, regardless of the energy level.^[106]

In a study conducted by Tang et al., the performances of different passivation strategies employing trometamol for solar devices were compared. The study investigated the effectiveness of bulk passivation, top-layer passivation, and dual passivation, individually. The device subjected to dual passivation exhibited a significantly higher efficiency compared with the devices subjected to top-layer passivation (18.66%) and bulk passivation (18.35%). Specifically, the dual passivation device achieved a PCE

of 19.17%. Therefore, the combination of bulk passivation and top-layer passivation using trometamol proved to be a more effective approach, resulting in improved PCEs.^[107] Garai et al. used the 2-benzoylpyridine molecule for the top-layer, bottom-layer, and bulk perovskite passivations simultaneously. This technology effectively reduced the defects in the perovskite and significantly improved the device stability.^[108] Since defects in perovskite do not occur only in certain parts but are generated simultaneously everywhere, passivating perovskite through double or triple passivation is essential for high-performance solar cells.

6. Defect Engineering of Lead-Free Perovskite

The toxic lead in perovskite is a factor hindering the commercialization of PSCs. Therefore, many researchers have attempted to develop solar cells using perovskite crystal films that do not use lead. Thus, various lead-free perovskites, such as tin-based perovskite, or double perovskite, have been proposed. The most representative is Sn perovskite, wherein Sn is used instead of Pb, and there are MASnI₃, FASnI₃, and CsSnI₃. However, the Sn–I bonds are easily broken by the highly energetic Sn 5s² states, leading to a low formation energy of the Sn vacancies (V_{Sn}). Additionally, the Sn 5s² states are not stable at high energies, allowing the easy oxidation of Sn (II) to Sn (IV), which induces deep traps, and nonradiative recombination.^[12]

The defects in most Sn-based perovskites mainly form shallow traps. However, halide interstitials and antisite substitutions, which occur at high densities, form deep traps, and induce non-radiative recombination. In particular, oxygen easily becomes superoxide (O₂⁻) and oxidizes Sn, promoting the formation of uncoordinated Sn. In addition, the reaction rate of SnI₂ with FAI or

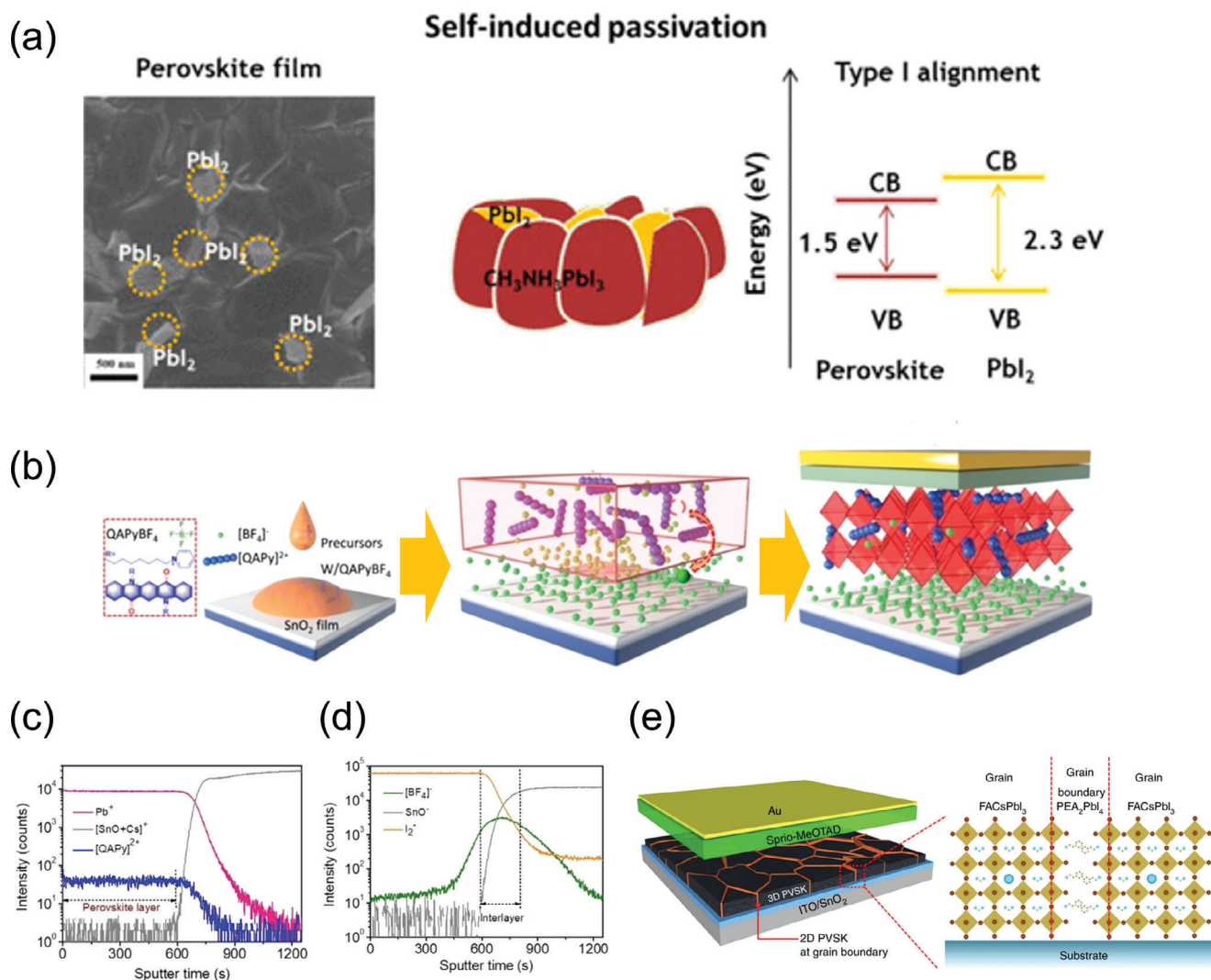


Figure 12. a) Schematic illustration of the self-induced passivation.^[96] b) Schematic illustration of the QAPyBF₄ passivation. c) Positive and d) negative polarity measurement of the TOF-SIMS depth profile of the perovskite with QAPyBF₄.^[102] e) Schematic illustration of the PSCs with PEA₂PbI₄ passivation.^[103]

MAI is higher than that of Pbl₂, resulting in rapid, and uncontrollable crystallization. Consequently, an inhomogeneous and defective film is produced, in contrast with the Pb-based perovskite film.^[109] Therefore, the aspect of defects in Sn-based perovskites requires special attention, and developing passivation strategies is a very important task.

Unlike Pb-based perovskites, Sn-based perovskites not only promote the coordination with Sn ion, but also suppress the oxidation of Sn. Tai et al. attempted to control Sn oxidation and defect formation by introducing a potassium salt of hydroquinone sulfonic acid (KHQA) into a Sn-based PSC.^[110] The hydroxy group (–OH) of KHQA inhibits the oxidation of Sn, and the sulfonate group (SO₃⁻) coordinates with Sn²⁺. Resultantly, the Sn-based PSCs incorporated with KHQA exhibited excellent oxidation stability and a significantly increased V_{OC}. The aspect of defects in Sn-based perovskites is more complicated than that in Pb-based perovskites because the oxidation of Sn must be considered. Therefore, further research is required to develop inno-

vative defect engineering strategies specifically for Sn-based perovskites.

7. Defect Engineering for Large-Scale Perovskite Solar Cells

Beyond lab-scale PSCs, the development of large-scale PSCs is essential for the commercialization of PSCs. However, the current large-scale perovskites have various constraints. These limitations are mostly caused by intrinsic defects and interfacial defects, attributed to the structural disorder and defects existing at the grain boundaries. Therefore, controlling the defects will allow the formation of a uniform and large-scale perovskite film.^[111]

In the spin-coating method, which has been widely used for depositing lab-scale perovskite films, a nonvolatile polar aprotic solvent is used. Unfortunately, this solvent is not suitable for depositing large-scale perovskites because of its low evaporation

rate. Therefore, various large-scale perovskite-deposition methods have been proposed, such as blade coating, slot-die coating, inkjet printing, and bar coating. Extensive research on methods for obtaining highly crystalline and less-defective films using each method is indispensable. Lee et al. demonstrated the effect of guanidinium iodide (GAI) additive for D-bar-coated PSCs. GAI effectively passivated the perovskite and improved the charge carrier lifetime. Consequently, a large-scale PSC ($4.8 \times 9.6 \text{ cm}^2$) with an efficiency of 19.44% was obtained.^[112]

8. Summary and perspective

In this review, we have discussed the defects generated in perovskites, their effects on PSCs, and various passivation strategies. Perovskite generally forms a polycrystalline film, and various defects are formed at the grain boundaries between these crystals because of the many dangling bonds. In addition, perovskite easily generates point defects, attributed to its ionic characteristics. These point defects induce ion migration, which leads to additional defect generation and stability inhibition. Notably, perovskite is defect tolerant, and shallow traps are primarily generated rather than deep traps. However, these shallow traps can transform into deep traps, highlighting the importance of addressing defects. Therefore, controlling the defects of PSCs through defect passivation plays a large role in improving the efficiency and stability of the devices.

Defects in PSCs show different characteristics at different locations; therefore, it is very important to identify these defect characteristics and implement tailored passivation strategies for each interface and location. We have expounded on the defects occurring at different locations of PSCs and on various passivation methods for treating each defect.

For top-layer passivation, effective approaches have involved using Lewis acids or bases, ionic compounds, 2D perovskites, and dopant-free HTL. On the other hand, bottom-layer passivation methods encompass Lewis acids or bases, ionic compounds, modified metal oxide ETLs, SAM molecules, and zwitterions. Meanwhile, passivation techniques targeting grain boundaries include strategies such as Lewis acids or bases, ionic compounds, self-induced passivation methods, doping cations or anions, and quasi-2D perovskite formation. We believe that this comprehensive elucidation of defect-engineering methods will significantly propel advancements in PSC research. We believe that this elucidation of defect-engineering methods will promote the advancement of PSC research.

Passivation will play a fundamental role in solving the current, significant challenges hindering the commercialization of PSCs. Firstly, in terms of efficiency, passivation has played a more important role than any other strategy, and we believe that it will continue to do so in the future. Notably, the V_{OC} values of existing PSCs are significantly different from the current S-Q limit, which is attributed to the nonradiative recombination induced by defects. Therefore, identifying the most optimal passivation method through systematic research is necessary. Machine learning presents an opportunity to augment the success of these systematic studies. By amalgamating machine learning with experimental approaches, comprehensive investigations into passivation's theoretical and practical facets become feasible. A recent attempt by Xu et al. to passivate perovskite defects

using a pseudo-halide anion serves as an exemplar.^[113] To discover the optimal pseudo-halide anion, density functional theory (DFT) calculations were systematically conducted via machine learning to screen suitable candidates. Subsequently, the introduced pseudo-halide anion underwent rigorous characterization to identify the optimal passivation material. This experimental methodology significantly diminishes trial and error, establishing a seamless connection between experimental theory and results.^[113]

Dual passivation or triple passivation is considered very effective in improving the efficiency and stability of PSCs because it simultaneously passivates the defects present in each site of the perovskite. However, the research on dual passivation and triple passivation is limited. Although this research area is quite complicated, particularly in terms of integrating each passivation technology, it is considered essential for PSC commercialization.

It is very efficient in process to passivate defects existing on the surface of perovskite using HTL or ETL, which has a functional group capable of passivation.^[114] This approach offers an advantage because it does not require an additional passivation process. Such HTL or ETL can prevent peeling from the surface of perovskite through interaction with the perovskite surface, reduce recombination that occurs on the surface, and make extraction occur easily. However, in the process of utilizing HTL and ETL for the passivation of surface-based perovskite defects, careful consideration of factors such as energy levels and carrier mobility becomes imperative.

Regarding the stability problem, passivation is the most essential solution. Defects accelerate ion migration, which can induce the transformation of shallow traps into deep traps, highlighting the need for passivation. In addition, defect passivation can improve phase stability and moisture stability when using hydrophobic molecules. The research on improving the stability of PSCs using defect engineering is in progress, and there are still many shortcomings in systematic stability testing using ISOS protocols. Therefore, many studies on defect engineering and stability should be conducted using the ISOS protocol to consider various stresses that may occur during the operation of PSCs.

Further research is required to explore the approaches for mitigating aging. The current passivation method, which relies on simple chemical interactions, is inadequate for preventing defects that arise from subsequent aging stress. Therefore, it is believed that the effective control of defects can be achieved by applying a method to reduce aging stress, such as heat management during device operation or UV light blocking, in conjunction with chemical passivation.

For the lead toxicity problem, considerable progress has been achieved with the introduction of Sn-based and double perovskites. However, the most representative lead-free perovskite, Sn-based perovskite, also has degradation and defects issues, and the oxidation problem of Sn complicates the defect engineering process of Sn-based perovskites. Certainly, the various defect-passivation strategies for Pb-based perovskite are still useful for Sn-based perovskite. However, since the oxidation problem of Sn has not been addressed, a passivation strategy unique to Sn-based perovskite is required. Predictably, the combination of machine learning and experiments will also advance this research area.

Defects are a fatal limitation even in large-area perovskite processing. During large-area processing, it is difficult to form a

uniform film by centrifugal force, and the nonuniform nanoscale phase impurity acts as a trap that promotes nonradiative recombination. Therefore, various deposition methods, such as blade coating, slot-die coating, and inkjet coating, rather than spin coating, are being explored. However, these methods are also limited by defect formation. Since the perovskite formed by these methods exhibits different characteristics from those formed by conventional methods, the defect characteristics are also different. Therefore, it is necessary to identify the most efficient optimization method by studying the defect characteristics for each deposition method.

To date, numerous papers have delved into perovskite passivation, yet none of which have conclusively determined the most optimal material or method. Extensive research is still required to identify the most effective passivation materials and techniques. Achieving this necessitates the establishment of systematic design principles derived from theoretical insights, coupled with further research. Integrating theoretical knowledge with machine learning is crucial in discerning the most optimal passivation materials and methods. Subsequent validation through experiments and diverse analyses remains imperative. Although the research on defect passivation in PSCs is complex and challenging, the potential rewards are great. With continued research, it is possible to develop highly stable, efficient, and affordable PSCs. This would be a breakthrough in the field of solar systems and could help accelerate the adoption of solar energy globally.

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Conflict of Interest

The authors declare no conflict of interest.

Keywords

defect engineering, defect passivation, defects, perovskite, perovskite solar cells, stability

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